Project 2 – Wireshark: TCP **CS 438/538, Money**

Gavin Money, 127.0.0.1:6789

2. A first look at the captured trace

	IP and TCP port# of Client		IP and TCP port# of the server
Q1	127.0.0.1:60708	Q2	127.0.0.1:6789

3. TCP Basics.

Question 4-6 (WS hints are given in the table)

	Sequence # in segment type	ACK Sequence # in segment type	What are in header to identify a segment type (SYN or SYN ACK)
Q3	SYN relative: 0; raw: 1242696871	relative: 0; raw: 0	SYN: Flags 0x002
Q4	SYNACK relative: 0; raw: 2015869677	SYNACK relative: 1; raw: 1242696872	SYNACK: Flags 0x012
Q5	HTTP GET relative: 1; raw: 1242696872	relative: 1; raw: 2015869678	Flags 0x018

Questions 7 and 8.

Order of segments	length	Receiver window	Any lack of buffer
#1	59	64256	No
# 2	65280	64256	Yes
# 3	256	64256	No
# 4	65495	64256	No (I'm positive this did not throw "window full")
# 5	4001	64256	No
# 6	65495	64256	No (I'm positive this did not throw "window full")

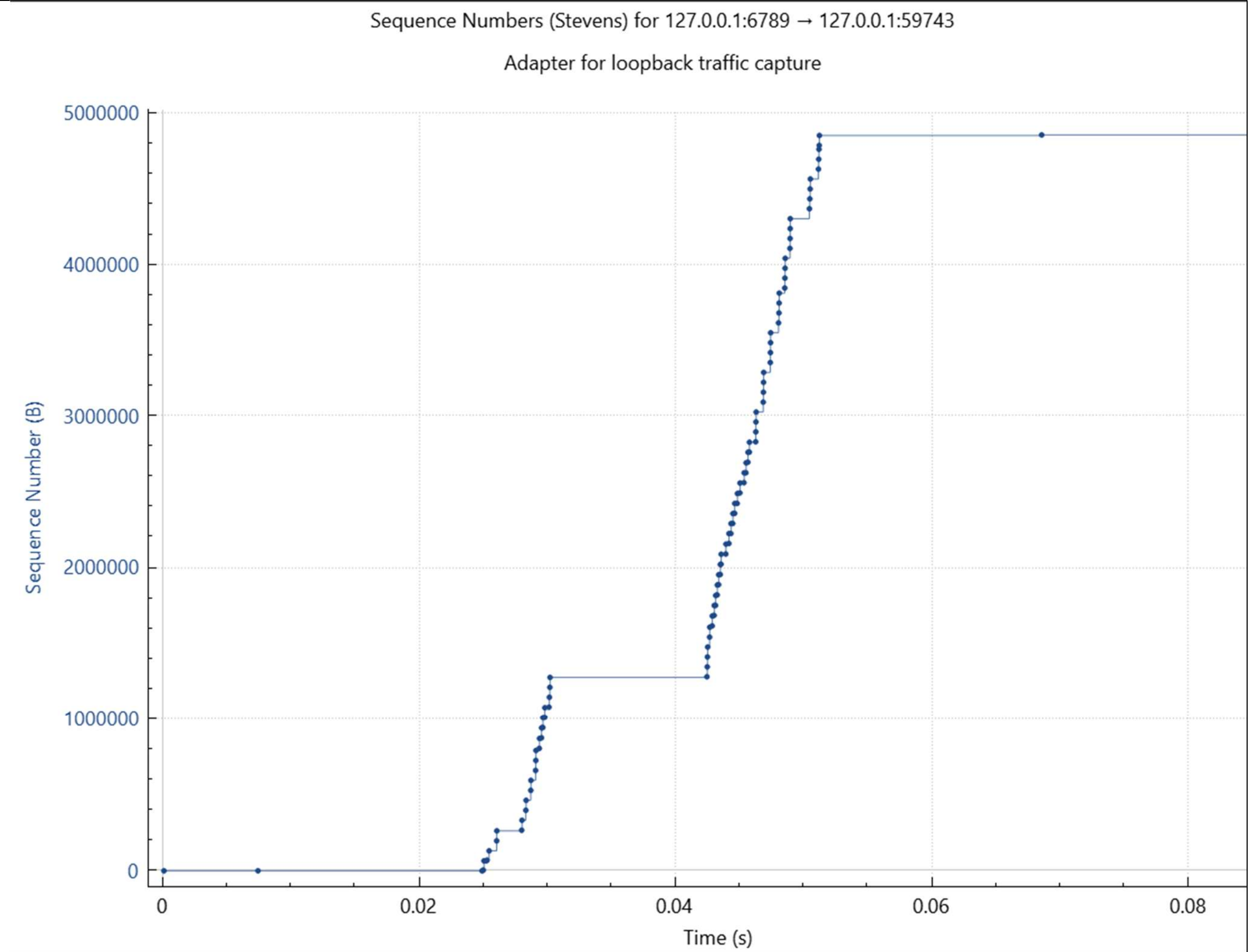
Questions 9 -10.

	value	explanations
Q9	No	Buffer (Packets are reconstructed later, after handling).
Q10	All data	No retransmissions needed.

4. TCP congestion control in action

Questions 12 -13. Use server V2 and *tree.JPEG*.

Q12 Insert the Time Sequence Graph below



Q13 Explanation.
The period is related to the roundtrip time, as the rise is likely acknowledgement packets and the flat parts are likely data transmission.

5. Additional: comparisons of response time and explanation

I combined the two tables. To gather the results, I looked at the last packet sent for each, and got the "timestamp since first frame in this TCP stream." V1 is understandably the slowest because it sends byte by byte. V2 is faster, but slower than V3 because it uses strictly TCP. V3 is the fastest because it has the ability to utilize UDP. Retrieving tree takes longer than CahabaLily for V2 because the file size is larger and more data needs to be sent over TCP.

V1 CahabaLily	**0.387 sec**
V2 CahabaLily	**0.012 sec**
V3 CahabaLily	**0.0281 sec**
V2 Tree	**0.06505 sec**